

Fontys HBO-ICT

Snapchat Hacking and Masking - Technical Manual

On behalf of the Rotterdam Police

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Version 1

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Version Control

Version	Date	Changelog
1	19/1/2026	First version of the document
2	22/1/2026	Adding Platform Tools to path troubleshooting

1. Introduction

The purpose of this document is to describe how all parts of the project are set up. This means setting up all requirements on the mobile phone, as well as installing the GUI with which all media is managed.

The GUI is both installable on Windows and Linux. Mobile devices tested and confirmed to be working are the following:

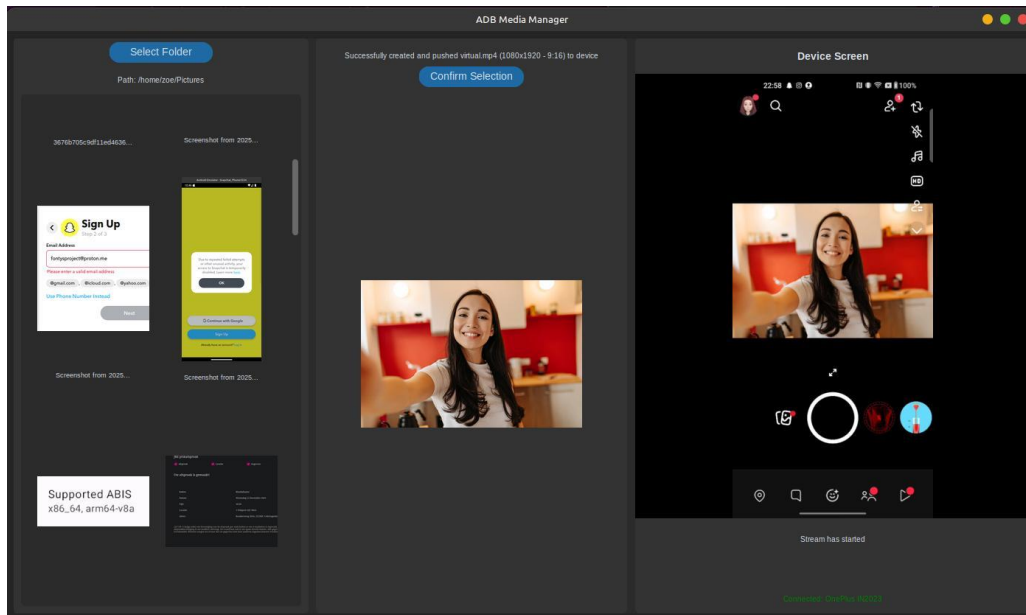
Name	Model	Operating System	Magisk	Snapchat Version
OnePlus 8 Pro	IN2023	Oxygen OS13.1 (Android 13)	Version 29	13.75.0.45 (Latest)
OnePlus Nord CE5	CPH2719	Oxygen OS16.0.1 (Android 16)	Version 30.6	13.75.0.45 (Latest)

When choosing a target device to utilize in this process, it is important to choose an Android mobile device, with a bootloader that can be unlocked. For example, a Samsung Galaxy S22 that was manufactured for the US has no option to unlock the bootloader, whereas the version that was manufactures for outside the US and Canada can still be unlocked (Vagrantmuse, 2022).

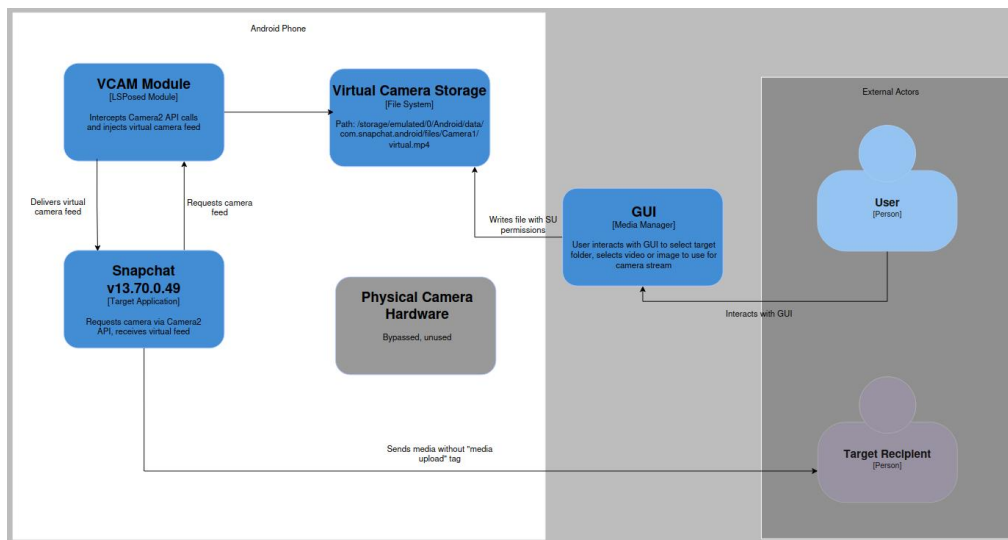
Research needs to be done prior to selecting a device. Usually, forums such as “XDA”, “Reddit”, or contributors on Github have a list of “rootable” devices (Zenfyrddev, n.d.). The difference not only lies between phone model, but often also in Android version. Always look up the specific model of phone used before purchase/updating. There are instances of phones that support Android Verified Boot keys, which allow one to run a custom OS with a locked bootloader, but those instances are rare (Chenxiaolong, 2024). More details can be found in the final advice document.

2. Current environment

1. GUI with integrated phone screen copy



2. Diagram (Simplified/Not technically accurate)



For the full details, please refer to documentation provided in the reading guide.

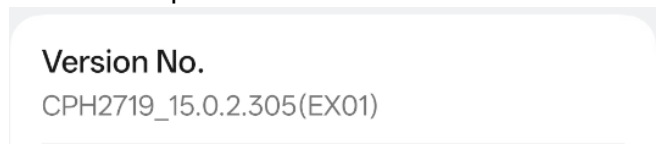
3. Installation Guide of Phone Environment

It is impossible to create a general rooting guide, as each device is rooted slightly differently. Consult the “XDA” forum (Jonker, n.d.) for instructions per phone, as well as tutorials on YouTube. Prefer to opt for tutorials that mention the keyword “TWRP”, as that allows for backup of the image created, and easy duplication of mentioned image on an identical device.

In this guide, a OnePlus Nord CE5 is used. It has also been tested on a OnePlus 8 Pro, with almost identical instructions. Should you choose a different phone brand, please root it accordingly and follow along from step 23. It is strongly advised once again to choose a brand based on the list in the sources, (Zenfyrdev, n.d.). Improper following of instructions can cause a “bricked” device, meaning, making it unusable.

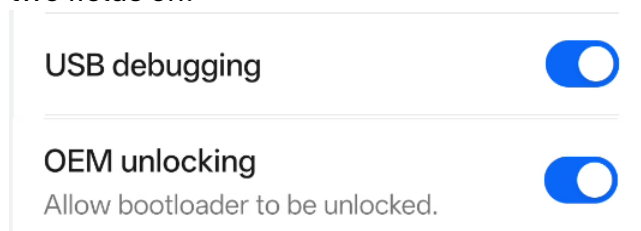
1. Enable developer settings

Go to Settings>About Device>Version and tap 7 times on “Version No.”, until you see ‘Developer Mode Enabled’



2. Enable USB debugging and OEM unlocking

Go to Settings>System & Update>Developer Options, and toggle the following two fields on:



3. Install USB drivers on computer

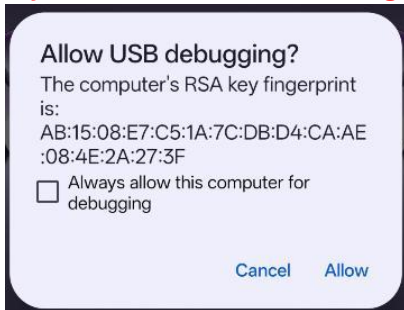
<https://developer.android.com/studio/run/win-usb>
<https://developer.android.com/tools/releases/platform-tools>

Follow the instructions in the above links if help is needed with installation.

4. Connect phone with USB-C cable to the computer

Unlock the phone screen and press allow in the dialog box:

If you don't see the following dialog box, re-plug the cable.



5. Ensure ADB recognizes device

In your command prompt¹, type “adb devices” and ensure the listed device is the one you’re trying to modify.

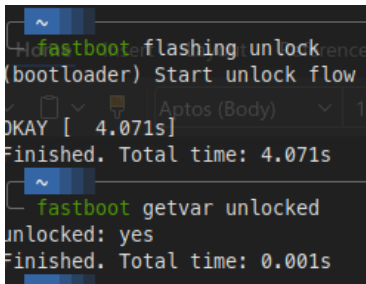
6. In your command prompt¹, type “adb reboot bootloader”

The device will reboot into fastboot mode.

7. In your command prompt¹, type “fastboot flashing unlock”

On the mobile device, press the volume button up to accept the unlocking of the bootloader.

After that, to confirm, in your command prompt¹ type “fastboot getvar unlocked”. You should see the following:



If not, repeat from step 5. The menu shows in tiny letters, and you have about 5 seconds to press the volume up button to lock your choice.

8. Reboot the phone

In your command prompt¹ type “fastboot reboot”

The phone will show an “orange state” warning, which is safe to ignore. This is because the device is now unlocked and not deemed safe. The phone will then reset itself and boot up to the welcome screen.

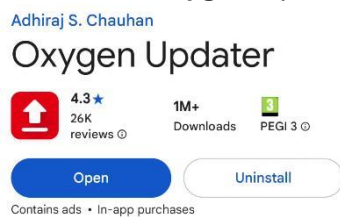
9. Connect phone to WiFi

Set up the phone accordingly, following the welcome guide. The most important step for now is connecting to the internet. In settings, go to “Wi-Fi” and connect to your network.

10. Download Oxygen Updater

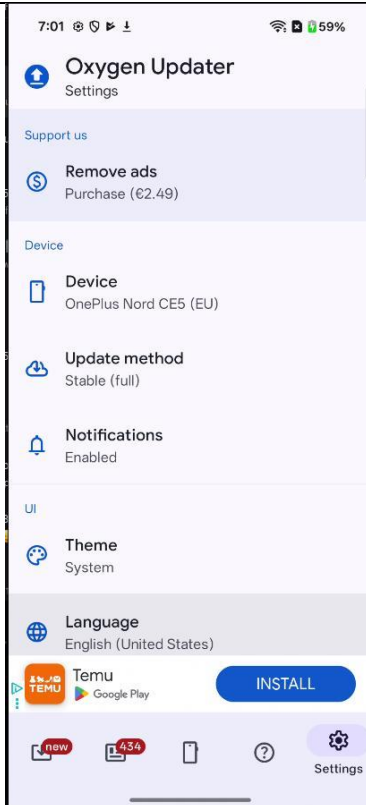
Open the Play Store. You will be prompted to sign in to your Google Account if you have not yet done so during the setup. Log in.

Search for Oxygen Updater within the Play Store and download the app.

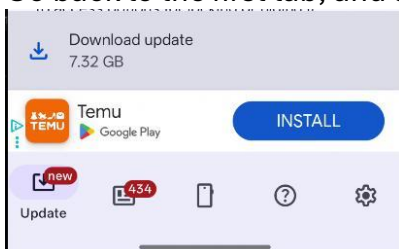


Download the latest full version

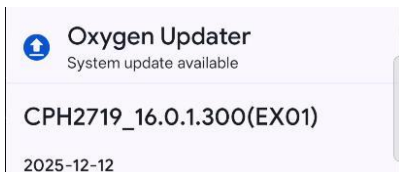
Go into the settings tab of the app and change the “Update method” to “Stable (full)”:



Go back to the first tab, and download the update:



Ensure that the version you are downloading on Oxygen Updater is the same version as the version currently on your phone:

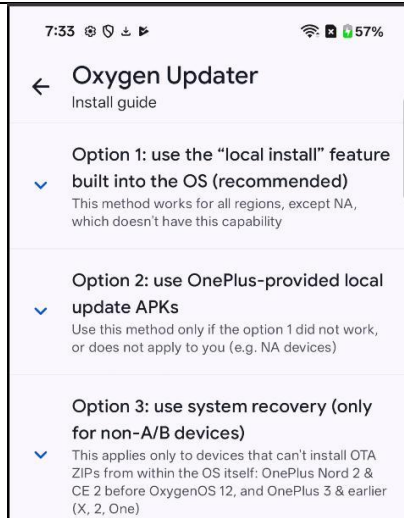


To check your phone's version, go to Settings>About device>Version:



As you see, there is a **mismatch** in this scenario. If you proceed without addressing this, you will make the device unusable (or very difficult to fix).

In this case, wait for the download to finish, and install the update on your phone. The "Install guide" will open automatically. Follow the instructions for Option 1:



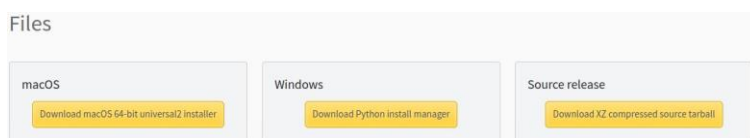
If "Local Install" does not show up, repeat Step 1, to gain back Developer Options. If "Local Install" is greyed out, you have to first install the update that is shown in settings. If after this "Local Install" is still greyed out, simply turn off the WiFi, close settings, and re-enter the Software Update menu.

Proceed with "Local Install" as shown in Oxygen Updater.

11. Install Python on your computer

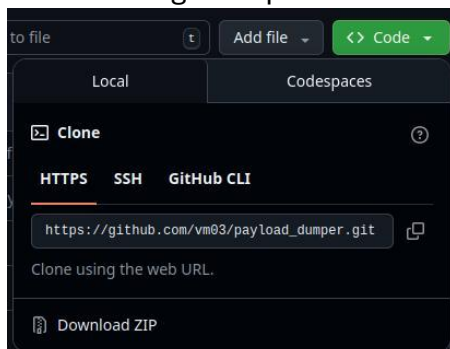
Download Python version [3.12.3](https://www.python.org/downloads/release/python-3123/) and follow the installer's instructions. If prompted to install "pip", agree.

<https://www.python.org/downloads/release/python-3123/>



12. Install Payload Dumper on your computer

Download https://github.com/vm03/payload_dumper, by clicking on "Code" and downloading the zip file:



In your command prompt¹ type “pip install protobuf” as it is needed by Payload Dumper.

Extract the Payload Dumper tool. In your command prompt¹ go to the location you extracted Payload Dumper. For example, if you extracted it in Downloads, type:

```
“cd %USERPROFILE%\Downloads\Payload_Dumper”
```

(Replace “Payload_Dumper” with the actual folder name if it is different)

Once you are in the correct directory, you can verify by typing: “dir”

You should see the Payload Dumper files listed before continuing.

Keep this command prompt¹ open, you will need it in a bit.

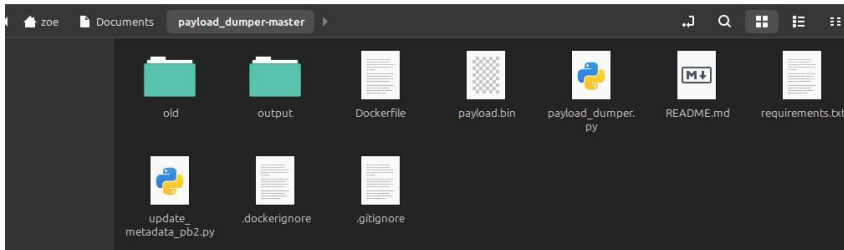
13. Copy the Oxygen OS update

Open a second command prompt¹ and in there, type “adb pull /sdcard/<file_name>.zip”

This will copy the zip file to your computer, in the current directory. Replace the name in <file_name> with the actual name, which can be found on your phone under My Files>Archives. It should be about 7GB.

14. Extract the zip file

Unzip the zip file. In the folder, you will find a file named “payload.bin”. Copy that file to the Payload Dumper folder.



15. Create virtual environment

In the open command prompt¹ that contains Payload Dumper, type “python3 -m venv venv”

Activate the virtual environment, by typing “venv\Scripts\activate”

Install the required dependencies, by typing “pip install -r requirements.txt”

A lot of dependencies will be installed. We are now ready to continue with Payload Dumper

```

Collecting typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (33 kB)
Downloading protobuf-6.33.4-cp39-abi3-manylinux2014_x86_64.whl (323 kB)
323.0/323.3 kB 11.1 MB/s eta 0:00:00
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading bsdiff4-1.2.6-cp312-cp312-manylinux_2_17_x86_64_manylinux2014_x86_64.whl (33 kB)
Downloading brotli-1.2.0-cp312-cp312-manylinux2014_x86_64_manylinux_2_17_x86_64.whl (1.4 MB)
1.4/1.4 MB 11.1 MB/s eta 0:00:00
Downloading zstandard-0.25.0-cp312-cp312-manylinux2014_x86_64_manylinux_2_17_x86_64.whl (5.5 MB)
5.5/5.5 MB 11.1 MB/s eta 0:00:00
Downloading fsspec-2026.1.0-py3-none-any.whl (201 kB)
201.0/201.0 kB 11.1 MB/s eta 0:00:00
Using cached requests-2.32.3-py3-none-any.whl (64 kB)
Downloading aiohttp-3.13.3-cp312-cp312-manylinux2014_x86_64_manylinux_2_17_x86_64_manylinux_2_28_x86_64.whl (1.8 MB)
1.8/1.8 MB 11.1 MB/s eta 0:00:00
Downloading aiohappyhelli-2.6.1-py3-none-any.whl (13 kB)
Downloading aiosignal-1.4.8-py3-none-any.whl (7.5 kB)
Downloading attrs-25.4.0-py3-none-any.whl (67 kB)
67.0/67.0 kB 11.1 MB/s eta 0:00:00
Downloading certifi-2026.1.4-py3-none-any.whl (152 kB)
152.0/152.0 kB 11.1 MB/s eta 0:00:00
Using cached charset_normalizer-3.4.4-cp312-cp312-manylinux2014_x86_64_manylinux_2_17_x86_64_manylinux_2_28_x86_64.whl (153 kB)
153.0/153.0 kB 11.1 MB/s eta 0:00:00
Downloading frozenlist-1.8.0-cp312-cp312-manylinux1_x86_64_manylinux_2_28_x86_64_manylinux_2_5_x86_64.whl (242 kB)
242.0/242.0 kB 11.1 MB/s eta 0:00:00
Using cached idna-3.11-py3-none-any.whl (71 kB)

```

16. Create partition image files

In the same command prompt¹, run the following command “python3 payload_dumper.py payload.bin”. This will create a lot of partition images, that will be placed in the “output” folder in the Payload Dumper folder.

17. Send init_boot.img back to the phone

From the same command prompt¹, go into the “output” folder, by typing “cd output”

To send the “init_boot.img” file to the phone, type “adb push init_boot.img /sdcard/”

Confirm if this was successful by typing “adb shell ls /sdcard/boot.img”

Once confirmed, you are ready to patch the image with Magisk.

18. Download and install Magisk

Download Magisk on your computer, from here:

<https://github.com/topjohnwu/Magisk/releases/tag/v30.6>

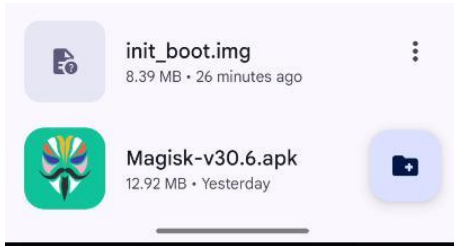
Scroll down to assets, and click on “Magisk-v30.6.apk” to download it

Assets 5				
app-debug.apk	sha256:5201ff9b1a53675ab3427ec827...	24.2 MB	Dec 2, 2025	
Magisk-v30.6.apk	sha256:f1ffc3c9a5614c251ba8bada39...	12.3 MB	Dec 2, 2025	
notes.md	sha256:8fabd1b038084454d899d08e...	817 Bytes	Dec 2, 2025	
Source code (zip)			Dec 2, 2025	
Source code (tar.gz)			Dec 2, 2025	

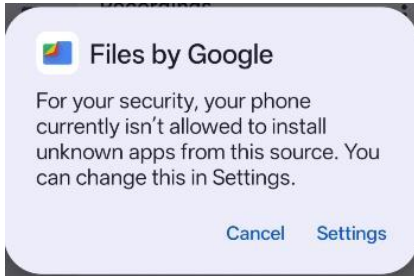
Copy the .apk to your phone. To do this, open a command prompt¹ and go to the directory you downloaded Magisk in. For Downloads, for example, go to “cd C:\Users\<YourUsername>\Downloads”

Once you are in the Downloads folder, you can run “adb push Magisk-v30.6.apk /sdcard/”

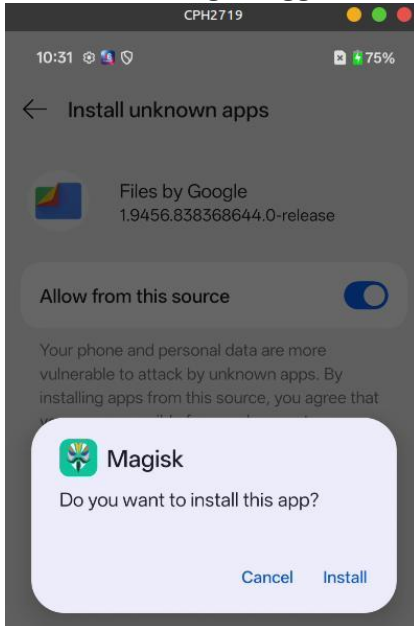
On your phone, in Files>Internal Storage, you should see the files you transferred:



Tap on Magisk to install it. You will receive this warning:

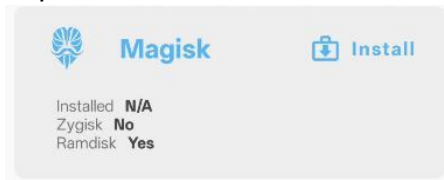


Click on Settings, toggle allow for this source, and install Magisk:

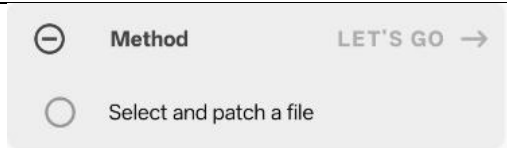


19. Patch init_boot.img

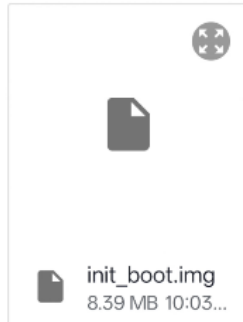
Tap “install”



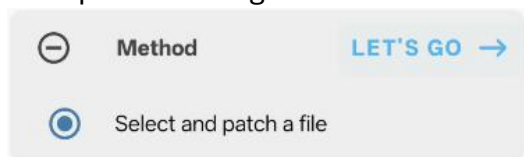
Click “Select and patch a file”



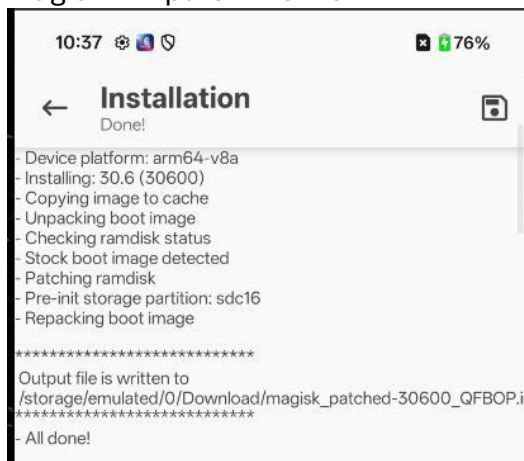
And find the “init_boot.img”. Select it.



Now press “Let’s go”



Magisk will patch the file:



20. Copy the patched file to your computer

In your command prompt¹, copy the patched image, by typing “adb pull /storage/emulated/0/Download/<name of patched magisk file, as seen in screenshot>.img

It will copy the folder/path your command prompt¹ is currently in. Rename this file on your computer to “mp.img”.

21. Root the device

To root the device, you will now use the patched image you created. In your

command prompt¹ type the following “adb reboot bootloader”. The device will go into the bootloader screen.

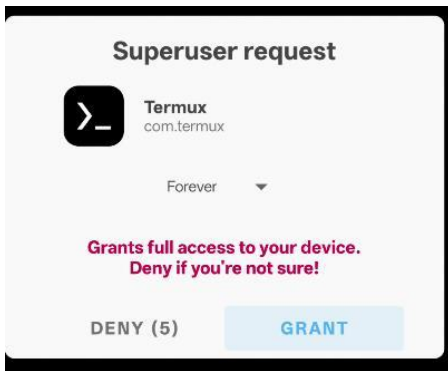
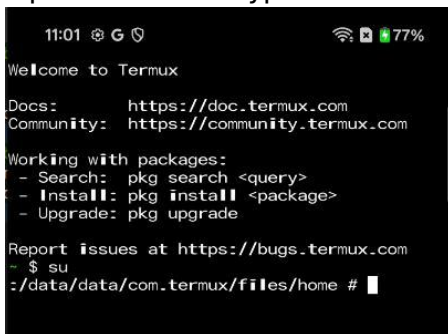
Then type “fastboot flash init_boot mp.img” Once that action is complete, type “fastboot reboot”, to boot into your now rooted phone.

22. Verify root

From the PlayStore, download “Termux”



Open Termux and type “su”:



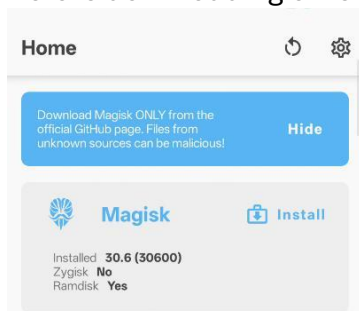
Click on “Grant”

You should be able to see Termux as a superuser now in Magisk:

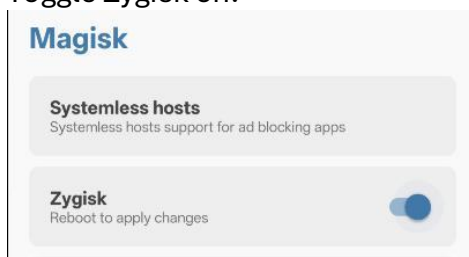


23. Enable Zygisk

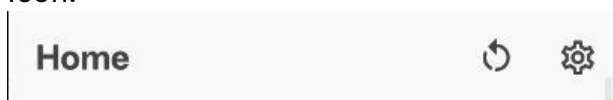
Before downloading other modules, enable Zygisk in Magisk settings:



Click on the Settings Icon,
Toggle Zygisk on:

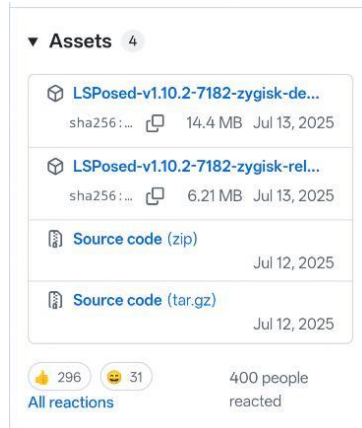


Go back to Magisk's home screen and reboot the device, the icon next to Settings icon:

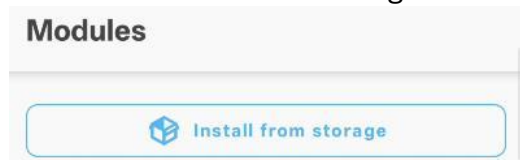


24. Download LSPosed

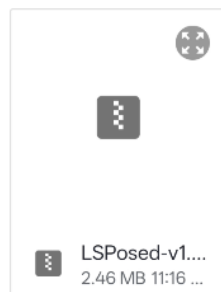
On the phone, browse to <https://github.com/JingMatrix/LSPosed/releases>
Scroll to the “Assets” section and download the 2nd file, by clicking on it.



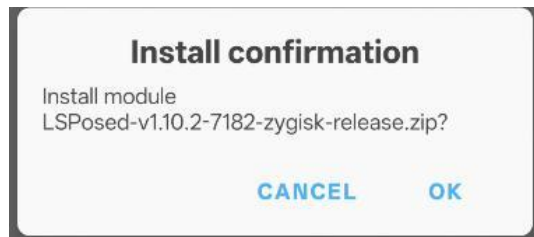
Do not open the file after it downloads. Instead, go to Magisk, click on “Modules”, and then “Install from storage”:



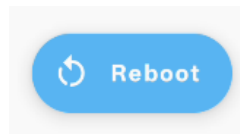
Find the file in the Downloads folder and tap it:



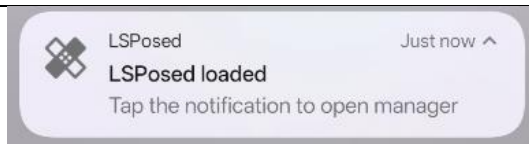
Confirm the installation:



Press “Reboot” once the installation is done:

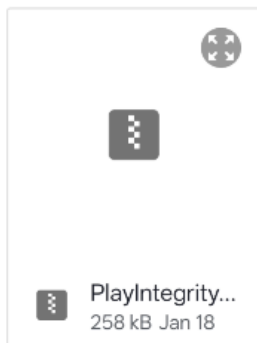


Once the phone starts up again, you should see that LSPosed is successfully loaded, by revealing your notifications:

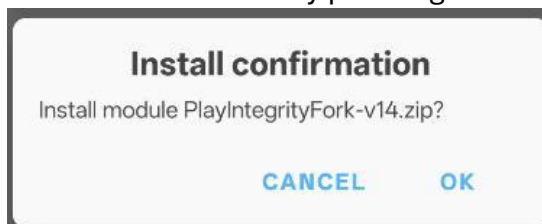


Do not open the files as they download, you will open them in Magisk.

Same steps as with LSPosed, go to “Modules” and “Install from storage”. Select Play Integrity Fork first:



Confirm installation by pressing “Ok”:



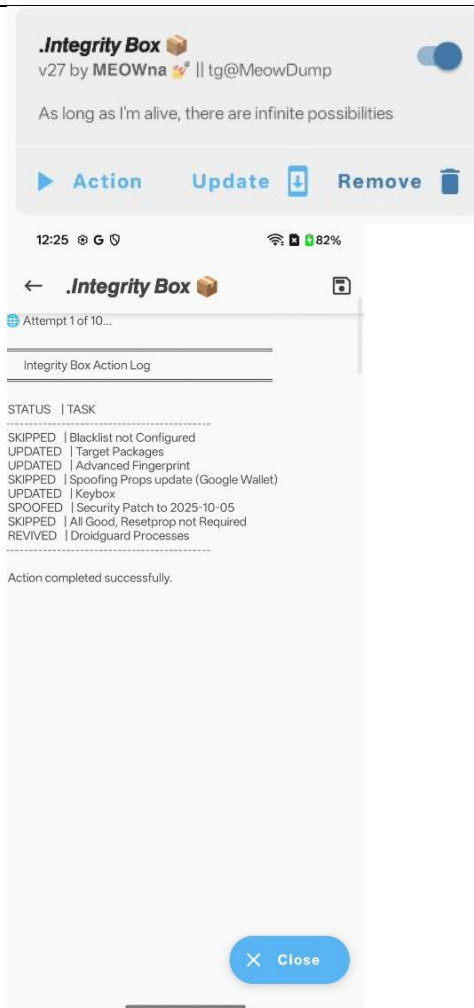
Do not reboot just yet, go back and repeat the process for Tricky Store, and finally Integrity Box.

After installing Integrity Box, your phone will open to a telegram link, safely ignore the webpage and go back to Magisk. Examine the log, and you should see “Installation Complete”.

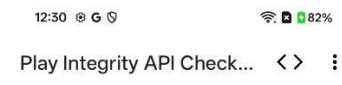
Reboot the device:



Once rebooted, go back to Magisk, Modules, and press the “Action” button for Integrity Box:



Re-run the integrity scan performed at the beginning of this step. If Device Integrity is not green, update Tricky Store, reboot, and rerun the “Action” for Integrity Box. Scan again.



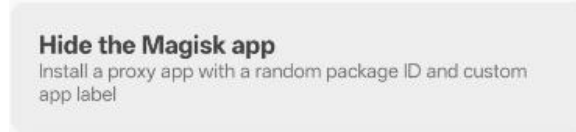
-  MEETS_BASIC_INTEGRITY
-  MEETS_DEVICE_INTEGRITY
-  MEETS_STRONG_INTEGRITY

CHECK

25. Hide Magisk

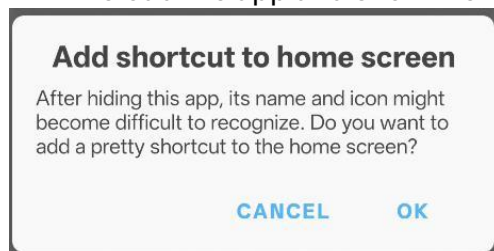
At this point you have all required modules on Magisk. From the home screen in Magisk, access Magisk's settings as before.

Scroll down and tap on "Hide the Magisk app":



The default name it will give itself is "Settings, this is okay. Tap "Ok".

It will reload the app and show the following dialog box:



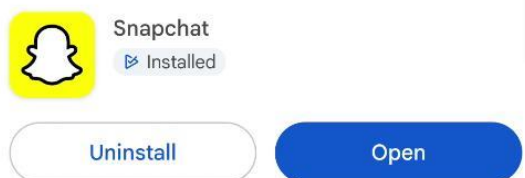
Press "Ok" and add it to the home screen. On the home screen however, remove the Magisk shortcut, and instead just keep this:



This is Magisk, but disguised.

26. Download Snapchat and sign in

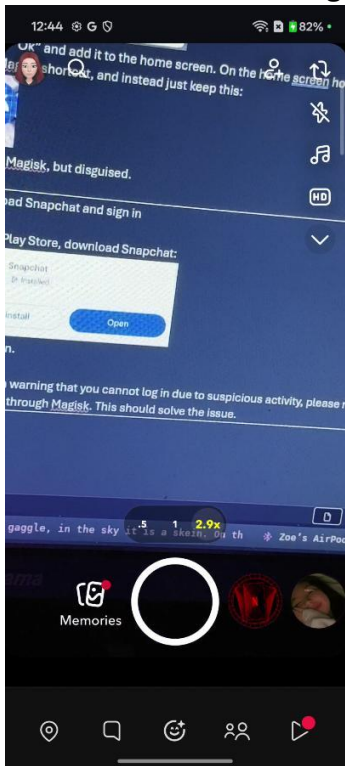
On the Play Store, download Snapchat:



And sign in.

If you get a warning that you cannot log in due to suspicious activity, please

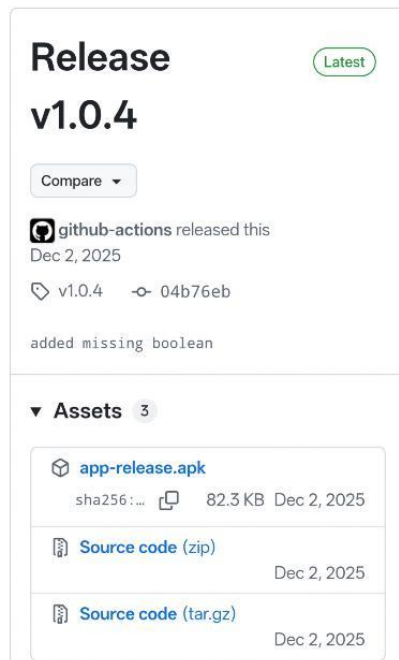
reboot the device through Magisk. This should solve the issue.



27. Load the Virtual Camera Module in LSPosed

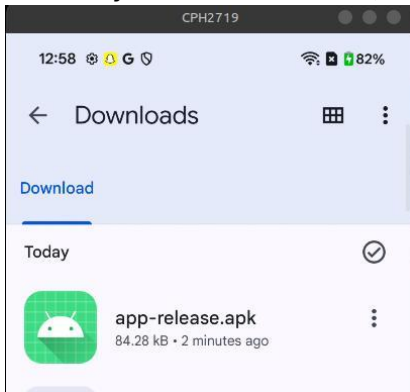
Download the latest version of the .apk file from

<https://github.com/CommitUninstall/com.example.vcam/releases>

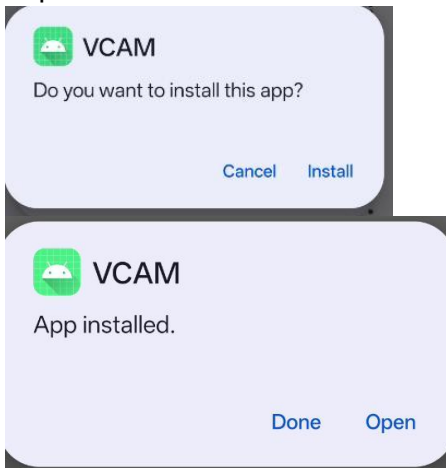


Click on “app-release.apk” to download it.

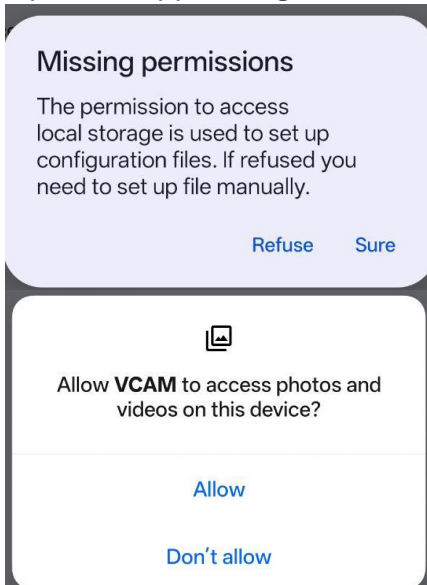
Go to My Files>Downloads, and tap on “app-release.apk” (My Files, not Files)



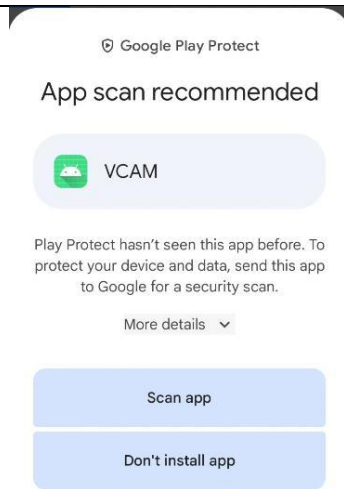
Tap “install”:



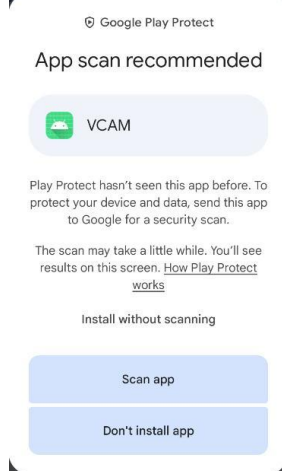
Open the app and agree to all dialogs you might see there.



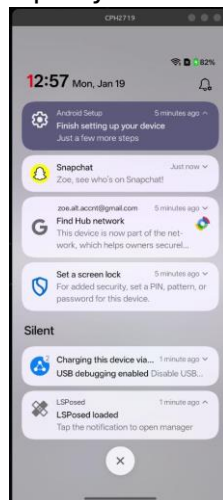
The following dialog might open during installation:



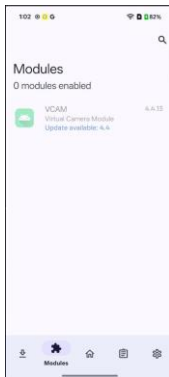
Press “More details” and “Install without scanning”



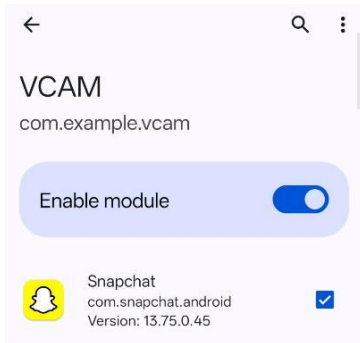
Open your notifications bar, tap on LSPosed:



Go to the “Modules” tab and click on “VCAM”:



Enable the module as well as Snapchat in the menu:



28. Open Snapchat

Open Snapchat, and you will notice that at this point, you see a notification near the bottom of your screen, stating that there is no video file found. You may now use the GUI to transfer your first video

29. Download the GUI

The GUI is privately managed on Github. The stakeholder of this project should have access to the Github page, or a local version of the project. If this is not the case, reach out to spyrouzoe@gmail.com or [CommitUninstall](#) on Github.

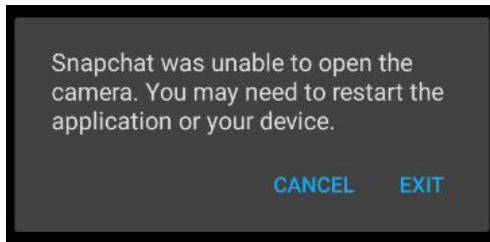
30. Launch the GUI

Right now, all that is left is to transfer the photo or video to the phone. Launch the GUI on your computer, and follow the steps, as shown in the User Guide

¹ The terminal can be opened by pressing “Win” + “R”, This opens a dialog, after entering “cmd” and pressing “Run” you should see the command prompt.

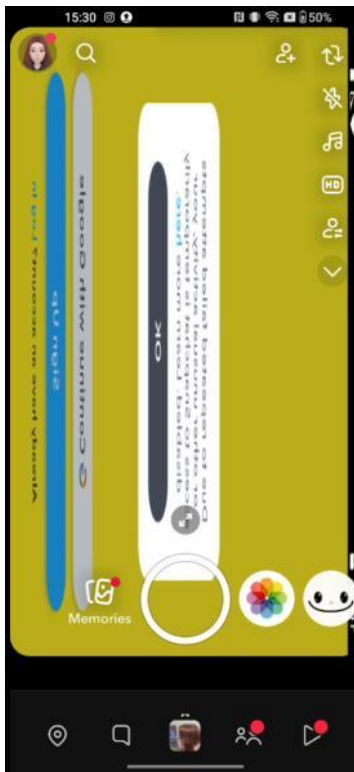
4. Troubleshooting and known issues

- When opening Snapchat, an error will show up if the app was fully closed before:



This is not an issue, just go to the home screen or drag up all active apps (do not fully close Snapchat), and then re-enter the application. This has to do with the Surface API and cannot be fixed, as it is a timing and attaching issue.

- While the LSPosed module works, there is a limitation when working with apps that use the “MediaToFrames” method, which use the JPEG format as opposed to YUV-420-888 for example. They do not receive the proper media rotation metadata, causing them to load into Snapchat rotated and squished:



The fix has been made on the GUI’s back-end, in the “image_to_video.py” file.

- Videos uploaded do not have sound. At this point there is no known solution.
- No 2-finger touch support on stream. Photos and videos need to be zoomed in on, on mobile device. This is a limitation of the GUI and further support for a zoom, or “pinch” function, would have to be integrated later, if needed.

- ADB command not found, yet Platform Tools installed? There are 2 options to solve this:
 1. Open a terminal in the extracted Platform Tools folder, and work from there
 2. Add Platform Tools to the Environmental Variables of Windows. To do this, look up “Environment Variables” on your computer. In the menu that opens, under “user Variables” (not “System Variables”), click “New”, and paste the folder path of your Platform Tools installation. Close any open command prompt windows and reopen them, to see if “adb” can be used now.

5. Update management

The phone should never be updated through the phone’s update manager. Any updates should preferably be ignored, to keep the stability of the environment, as major changes were not tested yet. Should you have to update, e.g. Snapchat does not support the current Android version anymore, consult the following documentation:

<https://topjohnwu.github.io/Magisk/ota.html> (OTA Upgrade Guides, n.d.).

Should you update through the phone’s update manager, you will most probably have to repeat all the above-mentioned steps.

6. References

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